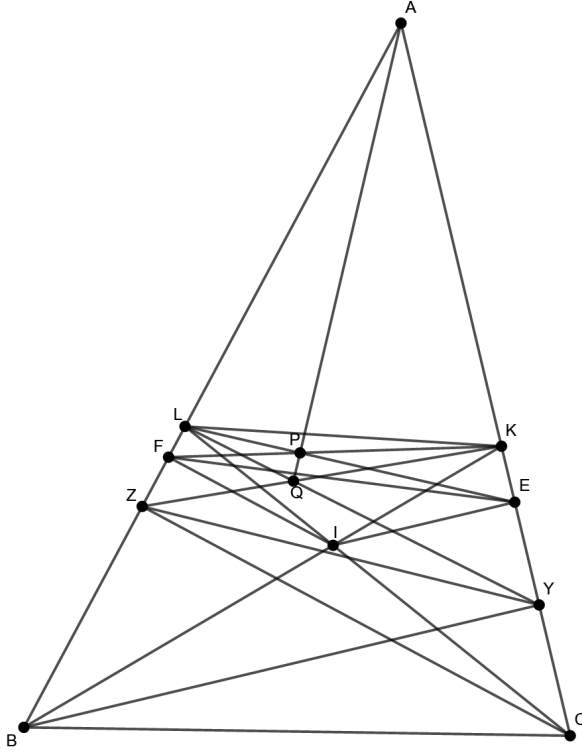


4.3.26



Let I be the incentre of $\triangle ABC$, P be the intersection of EL and FK , Q be the intersection of KZ and LY .

$$\begin{aligned}
 \frac{FP}{PK} \times \frac{KQ}{QZ} \times \frac{ZA}{AF} &= \frac{\triangle EFL}{\triangle EKL} \times \frac{\triangle KLY}{\triangle LYZ} \times \frac{ZA}{AF} \\
 &= \frac{\triangle EFL}{\triangle ELZ} \times \frac{\triangle ELZ}{\triangle LYZ} \times \frac{\triangle KLY}{\triangle EKL} \times \frac{ZA}{AF} \\
 &= \frac{FL}{LZ} \times \frac{AE}{AY} \times \frac{KY}{EK} \times \frac{ZA}{AF} \\
 &= \frac{IL}{CL} \times \frac{AE}{AF} \times \frac{AZ}{AY} \times \frac{BK}{IK} \\
 &= \frac{c}{a+b+c} \times \frac{b}{c} \times \frac{a+b+c}{b} \\
 &= 1
 \end{aligned}$$

By Menelaus theorem, A, P, Q are collinear.

By applying Desargues's theorem to $\triangle ELY$ and $\triangle FKZ$, KL, EF, YZ are concurrent.